COMP3329 Computer Game Design and Programming



Group 07

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Overview

Escapee is a 2D pixel art escape game. Along with engaging storytelling, the player has to interact with elements, solve puzzles and find items in order to escape.

The game consists of a map with various rooms to explore. When located in the corridors, movement and scenes follow 2D side scrolling. However within the rooms, this view is replaced by a top-down perspective.

Escapee creates an exciting atmosphere for players, challenging them to make their way out and win before time runs out. The setting of the scene is also very suggestive and familiar to many students within The Faculty of Engineering.

Scope of Project

Initial Game Idea Generation

We initially did not have any specific game idea, except for the general idea that the game should be an “Escape Room” game and in 2D, as we are all fans of the game “Ib”, a 2D horror adventure game released in 2012. This gave us the fundamental idea of the game play, which is to explore different rooms, interact with different items and escape the area. The actual game theme came to us when we joined a workshop at InnoWing and struggled to find the toilet.

Story Plot

The story is set in InnoWing. The player wakes up and finds themselves alone in the Dream Chamber. Their friend has taken away their student card, without which they cannot leave InnoWing. But that’s the least of their worries, as their bladder is getting full and they need to go to the toilet fast. They need to explore different rooms in InnoWing, solve puzzles, get different items, go to the toilet, and eventually escape InnoWing.

Feature Scope

During the planning phase of the game development, we came up with many ideas for different rooms and puzzles that could be included in the game. These ideas ranged from simple puzzles to complex rooms with their own mini games. However, due to the limited time, only the essential elements were implemented, such as the tutorial room, rooms for collectible items, and the toilet. Some of our more ambitious ideas, such as an AI patrolling security guard, unfortunately did not fit in our timeline and as such weren’t implemented.

The team had to make tough decisions on which ideas to prioritize and which to leave out due to time constraints. The *Future Plan* section illustrates all ideas and plans that were not implemented in the current version of our game.

As will be further discussed briefly, all pixel graphics, including the player's character, collectable and interactable objects, and scene backgrounds, were created by our groupmates. This allows for the game to have a consistent look and for it to be easily extendable in the future.

Finally, due to the limited time for the project, we were only able to create four endings for the game. These included one good ending and three bad endings. More details about the endings and their triggers will be explained in the *Scene* part under the *Game Element* section and the *Mechanism* part under the *Evaluation* section.

Graphics & Design

The game features its own original graphics. All items, characters, scenes and UI elements were designed by our team.

We made the decision to create everything by ourselves as a way to keep the game consistent and to allow for higher customisation. The created aesthetics, beyond being cohesive and fitting well together, give the game a distinctive look. All scenes were made especially to resemble InnoWing and to portray the feeling of the game.

Various popular pixel games were referenced as a source of inspiration but the biggest influence was the site of InnoWing itself, which our team visited on multiple occasions in order to include even the smallest of details.

Pixel art was utilised for all the graphics as a specific look for the game was being pursued (as mentioned inspired by other games such as “Ib”). Pixels made it easier to design a wider variety of elements along with more complicated scenes.

The aesthetics were also made to be fitting of the goals and tasks of the game with some scenes being darker and more eerie than others, while some more chaotic and confusing.

Coding & External Resources

We created the game using a variety of tools that aid the game’s development and aesthetics.

For game development, Unity 2D with C# scripts is used. GitHub is also used for version control which allows better collaboration, change tracking, and debugging. Other than that,

we used *Brackeys Tutorials* to learn techniques to efficiently use Unity and implement unique functions.

For the game aesthetics, we used *Piskel* and *Medibang Paint Pro* as our main digital art software. We also used *Fesliyan Studios* and *Pixabay* to source sound effects and music, specifically “Witness Testimony" by David Robson and "8 Bit Menu" by David Renda for the game’s background music.

Game Elements

Main Characters

Sprite	Character	Description
	Ione Win Chim	The central figure that players control. Ione unintentionally stayed up too late at InnoWing, woke up without their student card, and had an urgent need to use the restroom.
	Mr. S Guard	The InnoWing security guard. He has a 30% chance of appearing in the security area and triggering one of the bad endings of catching Ione.

Main Items

The player must acquire the following items in order to complete their goals and bring about the good ending. Each one requires solving a unique challenge in a separate room in the game.

Sprite	Items
	Toilet Paper Roll
	Key Card
	Plunger
	Soap
	Toilet Key

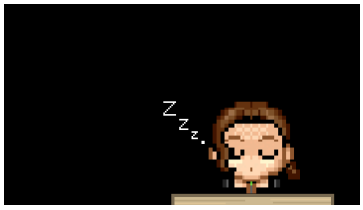
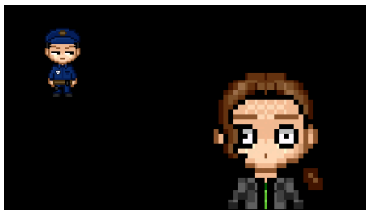
Scenes

While playing the game there are three types of scenes that can be encountered.

Views:

The first type are simple, zoomed in views portraying something that has happened in the game. These are used to present the beginning and endings (both good and bad) of the game and to show particular important moments in the game.

The table below contains all the main uses of this kind of scene.

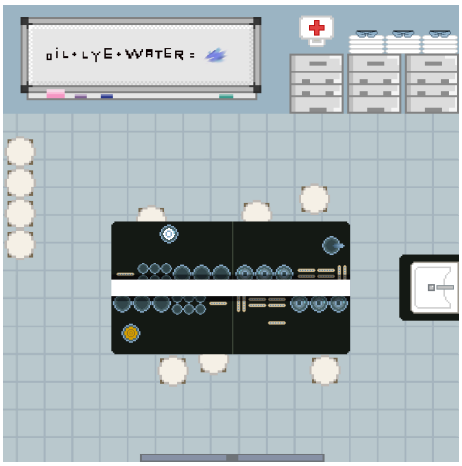
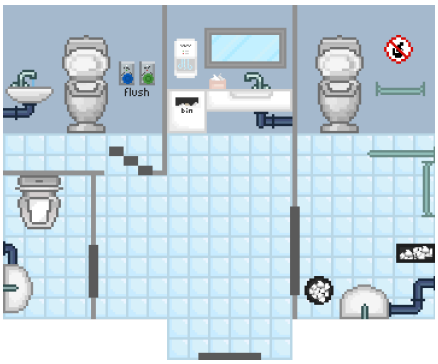
Sleeping Start Scene	Bad Ending 1 (“Accident”)	Bad Ending 2 (“Cannot Escape”)
		
Bad Ending 3 (“Caught by Security”)	Toilet Success Scene	Win Ending
		

Rooms:

The second types of scenes are views from within rooms. These areas offer a top-down perspective and within them the player can collect different items (by pressing the *‘spacebar’*), either by simply finding them or solving puzzles.

There are a total of six rooms.

Dream Chamber	Storage room
	

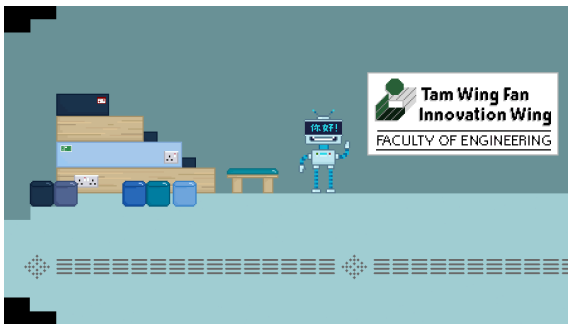
3D Printing Room	Lab
	
Security Room	**Toilet**
	

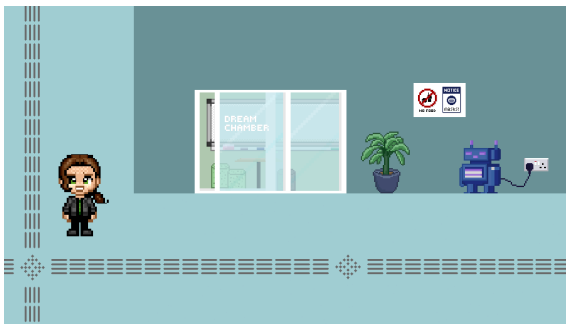
Corridors:

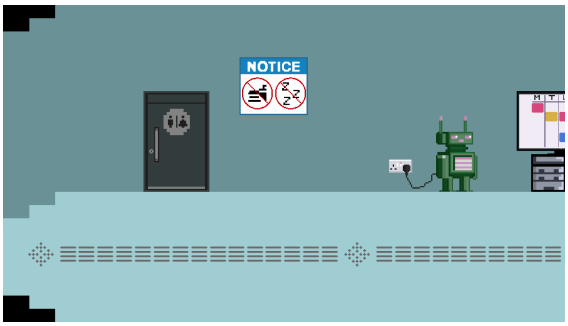
Lastly, there are corridors connecting the various rooms to construct a map of InnoWing. These areas present a 2D side scrolling view that allows players to explore the map and reach the various rooms through different doors. This is also where the Student Card can be randomly found.

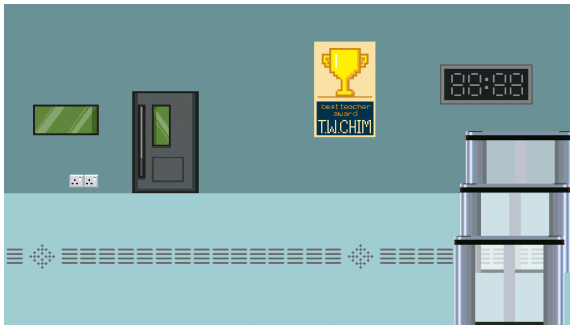
These scenes display a variety of elements and signs one could find in InnoWing.

Some corridor examples are reported below.







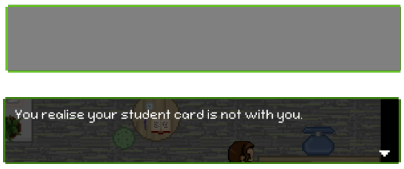


UI

The game's UI was designed to be simple and very easy for the users to understand.

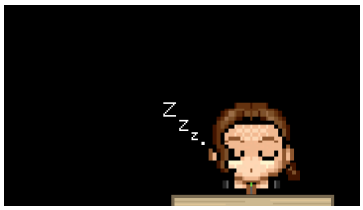
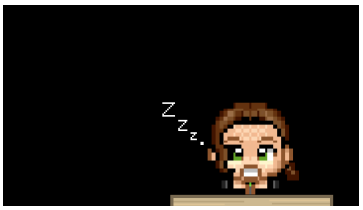
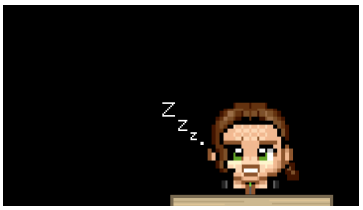
Four are the main elements that constitute the UI, they either display what is happening to users or allow for interaction. These are listed below:

	**Peedometer**	- represents the increase of the character's need to use the toilet as time passes. - starts from being empty and slowly fills up. - displayed on the top left corner of the screen
	**Inventory Box**	- exhibits the items that the player has collected and the ones still missing - elements appear just as an outline when not in the inventory box and show up in colour once they have been

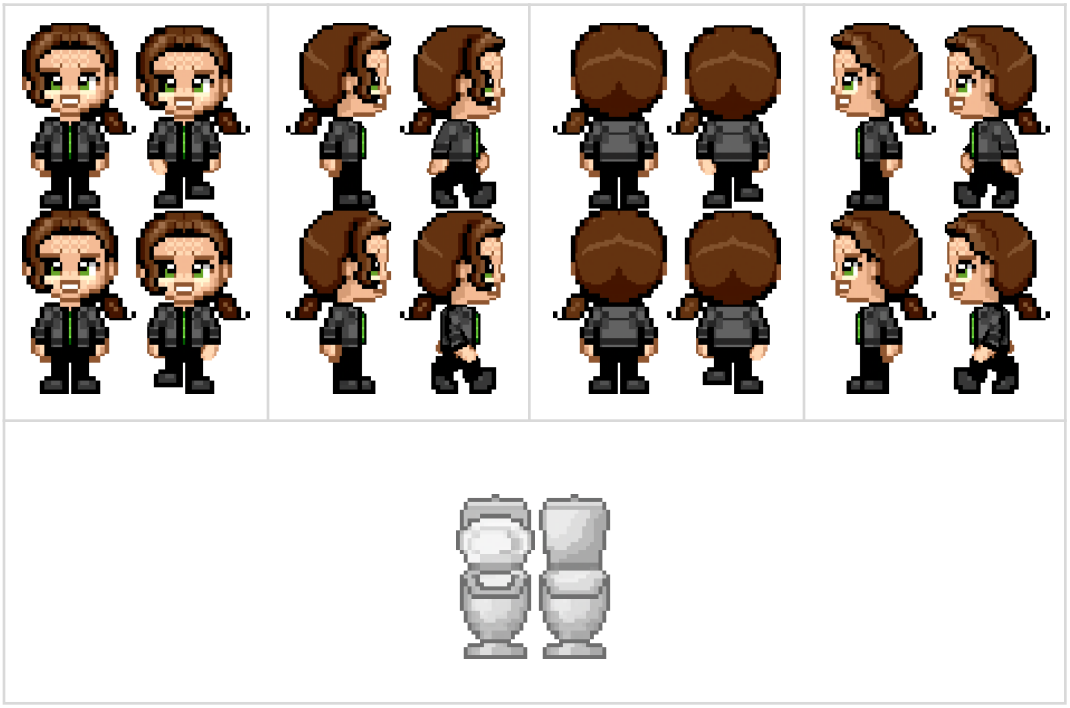
		retrieved. - displayed on the top right of the screen
	**Menu**	- features 2 separate buttons which allow either resuming the game or starting a new one - shown to player in the middle of the screen when *'esc'* button is pressed
	**Dialogue Box**	- shown when the player interacts with elements that have related dialogue - main game storytelling and communication tool - displayed at the bottom of the screen

Animation

Animations are included at important scenes to explain the plot and increase immersion in the game, notably, when the character wakes up in InnoWing at the beginning of the game, and successfully escapes InnoWing at the conclusion.

Additionally, there are unique object animations whenever the player interacts with the game, such as the character walking animation and the toilet lid.



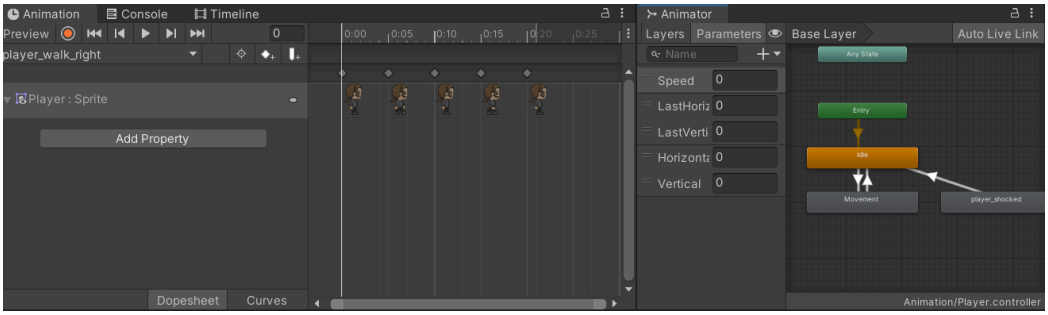
Evaluation: The Four Element Evaluation

Technology

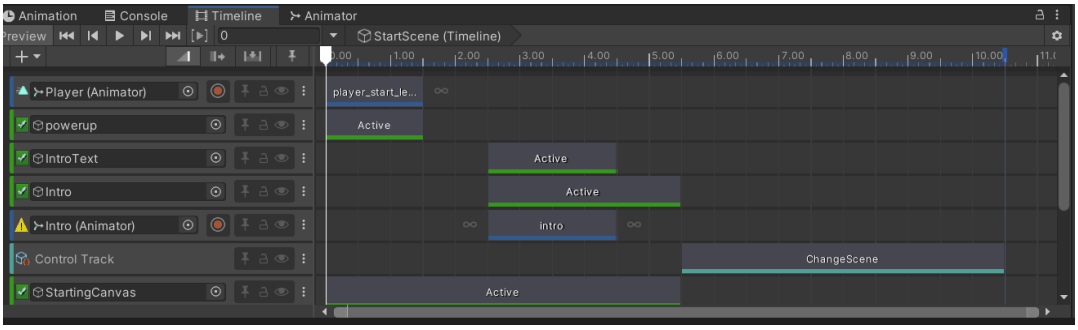
Escapee is made by Unity Game Engine (2D) which allows this game to be played on most modern hardware and operating systems, such as Windows, Mac OS, and Linux.

Other than the use of box colliders, scrolling screen, scene changing, etc, this game uses the timeline and animation tools of Unity.

Animation clips can be created by Unity's built-in animation tool (left) and can then be assigned to game objects using the Animator component. With the state machines (right) and C# scripts, complicated animation logic can be created, such as animation triggered by specific action and switching from one animation to another.



Meanwhile, the timeline tool helps manipulate animation clips, audio tracks, and script action with a simple drag-and-drop interface on a visual timeline. Cutscenes in this game, including the starting scene, ending scenes, and the toilet scene, are made by the timeline tool.



Mechanics

The player will control the character with "WASD" keys (or arrow keys) to navigate the 2D map of InnoWing's corridors and rooms, and "Space" key to interact with objects and enter/exit rooms. If the player needs to pause or restart the game, they can press the "Esc" key to open the menu. The "Esc" key will open the menu that offers the options to pause or restart the game, which the options can be navigated with the "WS" keys (or up and down arrow keys), and selecting an option with the "Space" key. The tutorial in the "Dream Chamber" room provides instructions for controlling the character.

As mentioned in the UI section under the Game Element part, the game features a Peedometer that records the player's pee-level and an inventory that shows the collected and missing objects.

The Peedometer acts as a countdown timer with a seven-minute time limit. It starts empty at the beginning of the game and gradually fills up over time during gameplay. If the player fails to go to the toilet and escape InnoWing before the Peedometer completely fills up, the "Accident" ending (bad ending 1) will be triggered.

The inventory shows all the objects needed for accessing the toilet and exiting the Innowing, At the beginning of the game, the icons in the inventory are just silhouettes. Once the player finds and collects an object, the corresponding icon will turn from black to colored. Access to the toilet requires the toilet paper, soap, plunger, and key. And exiting InnoWing requires the student card.

There are two random events in the game: the location of the student card and the chance of getting caught when visiting the security room. The student card will be randomly spawned on the floor of the corridors every time the player starts a new game. There will also be a 30% chance that the security guard will be in the security room, which triggers the "Caught by Security Guard" ending (bad ending 3).

In addition to the two ending triggers mentioned above, the game also has the "Win" ending (good ending) which is triggered when the player has accessed the toilet before the Peedometer is filled and has collected the student card to exit Innowing. The "Cannot Escape" ending (bad ending 2) is another one which is triggered when the player is able to access the toilet before filling up the Peedometer but does not have the student card to exit the building.

Story

Escapee emphasizes explorative gameplay, which means that the player is encouraged to explore and interact with every item in the game. Each item is designed to be interactable and a dialogue box with the item's description or the character's thoughts will pop up, engaging the player. This aspect of the game adds depth and complexity to the gameplay, allowing the player to immerse themselves in the world of *Escapee* and discover new things about the environment. This is especially important to puzzle-solving games, where paying attention to small details can be the key to progressing to the next step of the game.

The game also incorporates a time pressure element through the use of the Peedometer. Complementing the setting of the game, which is the player having their bladder full, this adds a sense of urgency to the gameplay and makes it more thrilling. Not only does the player have to explore the game world, but they must also do so efficiently to avoid running out of time.

The inclusion of multiple endings encourages the player to replay multiple times to see all the different game outcomes. This increases the replay value of the game, and also allows the player to make different choices based on their previous playthroughs.

Aesthetics

The atmosphere created by the combination of graphics, sound/visual effects and soundtrack allow the player to seamlessly immerse themselves in the game. As such a very engaging experience is created and it is possible to relate to the character in their mission to find a toilet.

The 2D pixel art graphics provide a simplified but seemingly technological and aesthetically pleasing representation of InnoWing. Meanwhile, the music builds suspense and helps give indication of what is going on in the game through well-placed sound effects.

As a whole, the blend of mechanics and graphics fit well together and create a very cohesive, consistent game that players can enjoy from a variety of aspects.

Future Plan

We plan to design and implement more rooms to extend the gameplay, for example, an AI Room, in which the player needs to defeat the AI robot in order to escape the room. Another room to be implemented is a “Doggo Room”, in which the player will encounter a robotic dog, Jet-propelled Autonomous Synthetic Pup Rover, or J.A.S.P.R., and can interact with it.

We also want to increase the difficulty of the game by introducing a security guard, who will patrol the corridor. The player must hide behind items or avoid the field of view of the guard, or otherwise, they will lose the game.

A save and load function of the game was planned to be incorporated. However, as the current version of the gameplay is only 7 minutes long, we have decided not to implement this function. As we develop other rooms and features in the future, with higher chances of losing the game, a longer gameplay in the post-toilet part of the game and more complex and extended storylines, we will want to include the save and load function to allow the player to return to their current progress, which can allow them to try for different stories. We will include savepoints in the form of card readers in the corridor.

Another plan we have for *Escapee* is to publish it on Steam and implement an achievement system for collecting different endings or triggering different easter eggs.

Conclusion

Escapee is a game successfully completed within three months for the COMP3329 Computer Game Design and Programming using a variety of technical development and aesthetic techniques. We received a lot of encouraging feedback not only on the consistency and level of detail in the game, but also the creativity of the theme. By watching other people play, we were also able to identify ways to further improve the gameplay aside from our future goals, such as raising the Peedometer period from 3 minutes to 7 minutes.

We are excited to launch the official release of *Escapee* in the coming summer of 2023.